Centralni limitni izrek

$X_1, X_2, \ldots$ neodvisne enako porazdeljene (NEP) slučajne spremenljivke z $E(X_1^2) < \infty$.

Označimo $S_n = X_1 + \ldots + X_n$

Za n dovolj velik je

$$P(a \leq S_n \leq b) \approx \Phi\left(\frac{b - nE(X_1)}{\sqrt{n\,var(X_1)}}\right) - \Phi\left(\frac{a - nE(X_1)}{\sqrt{n\,var(X_1)}}\right)$$

$$\Phi(x) = P(Z \leq x) \; ; \; Z \sim N(0,1)$$

Kocka vržemo 105-krat

a) Oceni verjetnost da pade < 350 pik.

X_i ... pike na i-ti kocki

X_i-ji so enako porazdeljeni $E(X_i) = 3{,}5$

$P(-\infty \leq S_{105} \leq 350) \approx$

$$E(X_i^2) = \sum_{i=0}^{6} \frac{1}{6} i^2 = \frac{1}{6} \cdot \frac{6 \cdot 7 \cdot 13}{6} = \frac{7 \cdot 13}{6}$$

$$var(X_i) = \frac{7^2}{4} + \frac{7 \cdot 13}{6^2} = \frac{35}{12}$$

$$P(S < 350) \approx \Phi\left(\frac{350 - 105 \cdot \frac{7}{2}}{\sqrt{105 \cdot \frac{35}{12}}}\right) = \Phi(-1) \approx 0{,}159\ldots$$

?? pri aproksimaciji je obaja enako

$$P(S \leq 349) = P(S < 350) = 0{,}145$$

Dejanska verjetnost: 0,152

$P(S < 349{,}5) = \ldots \approx 0{,}152$

V povprečju je srednja vrednost najboljša aproksimacija

b) Oceni vjetnost da 6 pade najveć 10-krat

I_i ... na i-tom mestu pade 6

$P(I_i = 1) = \frac{1}{6}$ $\quad I_i \sim Ber(\frac{1}{6})$

$var(I_i) = \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{6^2}$

$$P(S_{105} \leq 10) = P(S < 10{,}5) \approx \Phi\left(\frac{10 - \frac{105}{6}}{\sqrt{105 \cdot \frac{5}{36}}}\right) \approx 0{,}03$$

dejansko: 0,027

visoka relativna napaka

Problem: neenakomerna vjetnost: $\frac{1}{6}$ pri $\frac{5}{6}$

oziroma $\frac{1}{6}, \frac{1}{6}, \frac{1}{6} \ldots$

V sloveniji živi 51% žensk. Koliko klicev mora opraviti v Medianu, da ba vanketi sodelovalo vsaj pol žensk z vrjetnostjo vsaj 0,9?

X_i ... poklicana je ženska indikator

$X_i \sim Ber(p)$ $\quad p = 0{,}51$

$X_1 + \dots + X_n = S_n$

$P(S_n \geq \frac{n}{2}) \geq 0{,}9$

$\parallel$

$P(S_n \leq \frac{n}{2}) < 0{,}1$

$P(S_n \leq \frac{n}{2}) \approx \Phi\left(\frac{\frac{n}{2} - np}{\sqrt{np(1-p)}}\right) < 0{,}1$ $\quad = z$

$\Phi(z) < 0{,}1$

$z < -1{,}29$

$\sqrt{n}\left(\frac{1}{2} - p\right) < -1{,}29\sqrt{p(1-p)}$

$\sqrt{n} > \frac{1{,}29\sqrt{p(1-p)}}{-\left(\frac{1}{2} - p\right)}$

n mora biti vsaj [illegible]

mora biti vsaj [illegible]

3. $X_i \sim \text{Gama}(a, \lambda)$ neod

Izračunaj $P\left(X_1 + \dots + X_n \leq \frac{cna}{\lambda}\right)$

Momentna rodovna funkcija $M_X(t) = E(e^{tX})$

$$\boxed{\begin{aligned} &X, Y \text{ neodvisni:} \\ &M_{X+Y}(t) = E\left(e^{t(X+Y)}\right) = E\left(e^{tX} \cdot e^{tY}\right) \overset{\text{neodvisnost}}{=} E\left(e^{tX} e^{tY}\right) = \\ &= M_X(t) \cdot M_Y(t) \end{aligned}}$$

$$M_{X_1}(t) = E\left(e^{tX_1}\right) = \int_x e^{tx} \frac{\lambda^a x^{a-1}}{\Gamma(a)} e^{-\lambda x} \, dx =$$

$$\frac{\lambda^a}{\Gamma(a)} \int_{-\infty}^{\infty} e^{x(t-\lambda)} x^{a-1} \, dx \dots = \frac{1}{\left(1 - \frac{t}{\lambda}\right)^a}$$

$$\boxed{\text{Momentna rodovna funkcija enolično določa porazdelitev}}$$

$$M_{X_1 + X_2}(t) = \frac{1}{\left(1 - \frac{t}{\lambda}\right)^{2a}} = M_{\Gamma(2a, \lambda)}(t)$$

$$\Rightarrow \sim \Gamma(2a, \lambda) \quad \Longrightarrow S_n \sim \Gamma(na, \lambda)$$

$$P\left(S_n \leq \frac{cna}{\lambda}\right) = \int_{-\infty}^{\frac{cna}{\lambda}} \frac{\lambda^{na} x^{na-1} e^{-\lambda x}}{\Gamma(a)} \, dx =$$

$$= \int_0^{\frac{cna}{\lambda}} \frac{(\lambda x)^{na-1} \lambda e^{-\lambda x}}{\Gamma(a)} \, dx = \int_0^{cna} \frac{1}{\Gamma(a)} (u)^{na-1} e^{-\lambda u} \, du$$

DN: uporabi centralni limitni izrek

Ocene za napake pri CL1

Če imajo X_i tudi tretji moment

velja ocena

$$\sup_{x \in \mathbb{R}} \left| P(S_n \leq x) - \Phi\left(\frac{x - nE(X_1)}{\sqrt{n\,\mathrm{var}(X_1)}}\right) \right| \leq \frac{0{,}5583}{(n\,\mathrm{var}(X_1))^{\frac{3}{2}}} nE(|X_1 - E(X_1)|^3)$$

da se posplošiti za primer

ko $X_1 \dots X_n$ niso enako porazdeljeni:

① $X_1, X_2, \ldots$ neodv. in $X_k \sim \exp\left(\frac{1}{k}\right)$

Ali v tem primeru velja CLI?

$$S_n = \sum_{i=1}^{n} X_i \qquad E(X_k) = k \qquad var(X_k) = k^2$$

$$var(S_n) = \sum_{i=1}^{n} var(X_i) = \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

neodvisnost

$$E(|X_k - E(X_k)|^3)$$

$$\gamma_1^3 = E(|X_1 - E(X_1)|^3) = \frac{12}{e} - 2$$ ← naloga

$$X_k \overset{d}{=} k X_1$$

$$P(X_k > x) = e^{-\frac{1}{k}x} \qquad P(kX_1 > x) = P\left(X_1 > \frac{x}{k}\right) = e^{-\frac{1}{k}x}$$

← preživetvena funkcija

$$\Rightarrow E(|X_k - E(X_k)|^3) = E(|kX_1 - kE(X_1)|^3) = k \cdot \gamma_1^3$$

Oglejmo si

$$\frac{\sum_{k=1}^{n} E(|X_k - E(X_k)|^3)}{var(S_n)^{\frac{3}{2}}} = \frac{\gamma^3 (1 + \ldots + n^3)}{\left(\frac{n(n+2)(2n+1)}{6}\right)^{\frac{3}{2}}} = \frac{\gamma^3 \frac{n^2(n+1)^2}{4}}{\left(\frac{n(n+2)(2n+1)}{6}\right)^{\frac{3}{2}}}$$

$$= O\left(\frac{n^4}{n^{\frac{3}{2}}}\right) = O\left(\frac{1}{\sqrt{n}}\right) \longrightarrow 0$$

$\Rightarrow$ CLI velja

②

$p \in (0,1) \quad S \sim Bin(n,p)$

Ocen: $\left| P(S \leq x) - \Phi\left(\frac{x-np}{\sqrt{np(1-p)}}\right)\right| = *$

Obstajajo $X_1, \dots, X_n \sim Bernuli(p)$ neodvisne

$S = \sum X_i$

Poglejmo si

$$E(|X_1 - \underbrace{E(X_1)}_{p}|^3) = E(|X_1 - p|^3) =$$

$$= |1-p|^3 \cdot p + (1-p)|p^3| = p(1-p)((1-p)^2 + p^2) \leq p(1-p)$$

$$* \leq \frac{c\, np(1-p)}{(np(1-p))^{\frac{3}{2}}} = \frac{c}{\sqrt{np(1-p)}}$$

③

$X_1, \ldots, X_{100}$ neodv. enako porazdeljene

$X_1 \sim \begin{pmatrix} -1 & 1 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$

$$S = 10X_1 + X_2 + \ldots + X_{100}$$

Oceni verjetnost

$P(S < 41)$

$$\star = P(S < 41 \mid X_1 = -1)\underbrace{P(X_1 = -1)}_{=\frac{1}{3}} + P(S < 41 \mid X_1 = 0)\underbrace{P(X_1 = 0)}_{=\frac{2}{3}}$$

$$P(S < 41 \mid X_1 = -1) = P(\tilde{S} < 51)$$

$$P(S < 41 \mid X_1 = 1) = P(\tilde{S} < 31)$$

$$\star \approx \frac{1}{3}\Phi\left(\frac{50 - 95 \cdot \frac{1}{3}}{\sqrt{95 \cdot \frac{8}{9}}}\right) + \frac{2}{3}\Phi\left(\frac{30 - 33}{\sqrt{88}}\right) \approx 0{,}57\ldots$$

Točen rezultat $\approx 0{,}569$

Če bi uporabili CLI direktno na S bi dobili $\approx 0{,}6$

$X_1, X_2, \dots$ neodv $\quad X_n \sim \begin{pmatrix} -n & 0 & n \\ \frac{1}{2n^2} & 1-\frac{1}{n^2} & \frac{1}{2n^2} \end{pmatrix}$

Pokaži, da CLI ne velja

Recimo da velja CLI

Potem $\lim_{n\to\infty} P(X_1+\dots+X_n$

$$E(S_n) = \sum E(X_k) = 0$$

$$var(X_n) = E(X_k^2) = 1 \quad \Rightarrow var(S_n) = n$$

$$W_n = \frac{S_n - 0}{\sqrt{n}} \qquad P(W_n \in [-\tfrac{1}{2}, \tfrac{1}{2}]) = \Phi(\tfrac{1}{2}) - \Phi(-\tfrac{1}{2}) =$$

$$= 1 - 2\Phi(-\tfrac{1}{2}) \approx 0{,}38$$

$n \geq 4$

$$P(-\tfrac{1}{2} \leq W_n \leq \tfrac{1}{2}) = P(-\tfrac{1}{2} \leq \frac{S_n}{\sqrt{n}} \leq \tfrac{1}{2}) =$$

$$= P(\tfrac{\sqrt{n}}{2} \leq S_n \leq \tfrac{\sqrt{n}}{2}) \geq P(X_2 = X_3 = \dots = X_n = 0)$$

$$\vdots$$

$$\frac{n+1}{2^n} > \frac{1}{2} \quad \text{za } n \geq 4 \quad \#$$

Naj bodo $X_1, X_2, \ldots$ slučajne spremenljivke z vrednostmi sl. spr. $(X_n)_{n \in \mathbb{N}}$ konvergira proti sl. spr. X v porazdelitvi, če

$$\lim_{n \to \infty} F_{X_n}(x) = F_X(x) \quad \text{za } \forall x \in \mathbb{R}, \text{ kjer}$$

je F_X zvezna

$X_n = \frac{1}{n}$

Očitno bi želeli, da konvergira proti 0

$$F_{X_n}(x) = \begin{cases} 1 & x \geq \frac{1}{n} \\ 0 & \text{sicer} \end{cases}$$

Ogl

$$F_X(x) = \begin{cases} 1 & x \geq 0 \\ 0 & \text{sicer} \end{cases}$$

Ogl opazimo si tudi nekonvergenco

$F_X(0) = 1 \quad F_{X_n}(0) = 0 \quad \forall n \in \mathbb{N}$ *

Oglejmo si, da po definiciji velja pravzaprav kjer je konvergenca

Nej bo $x \neq 0$

$F_X(x) = 1$; če $x > 0$

$F_X(x) = 0$ če $x < 0$

$F_{X_n}(x) = 1$ ce $x > \frac{1}{n}$

$F_{X_n}(x) = 0$ ce $x \leq \frac{1}{n}$

$X_n \xrightarrow{d} X$, kjer $X \sim Pois(\lambda)$

$\lim F_{X_n}(x) = F_X(x)$, kjer je F_X zvezna

$$P(X=k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

$$\lim_{n\to\infty} \frac{P(X_n=k)}{P(X_n=k-1)}$$

$$\frac{P(X_n=k)}{P(X_n=k-1)} = \frac{\binom{n}{k} p_n^k (1-p_n)^{n-k}}{\binom{n}{k-1} p_n^{k-1} (1-p_n)^{n-k+1}} =$$

$$= \frac{n-k+1}{k} \frac{p_n}{1-p_n} \qquad n p_n \to \lambda \Rightarrow p_n \to 0$$

$$\lim * = \frac{np_n}{k(1-p_n)} + \frac{p_n(1-k)}{k(1-p_n)} \xrightarrow[n\to\infty]{} \frac{\lambda}{k}$$

$$P(X_n=k) = \frac{P(X_n=k)}{P(X_n=k-1)} \frac{P(X_n=k-1)}{\dots} \cdots \frac{P(X_n=1)}{P(X_n=0)} P(X_n=0)$$

$$= \prod_{i=0}^{k} \frac{\lambda}{i} (1-p_n)^n = \lambda (1-p_n)^k \prod_{i=0}^{k} \frac{1}{i} =$$

$$P(X_n=0) = (1-p_n)^n = (1-p_n)^{\frac{np_n}{p_n}} \Rightarrow e^{-\lambda}$$

$$= \frac{\lambda e^{-\lambda}}{k!}$$

$$\lim F_{X_n}(x) = \lim_{n\to\infty} \sum_{k=0}^{\lfloor x \rfloor} P(X_n=k) = \sum \frac{\lambda e^{-\lambda}}{k!} = F_X(x)$$

(M, d) metrický prostor

$\tilde{M} \subseteq M$

$\tilde{x}_1, \tilde{x}_2, \dots \tilde{x}$ sl. spr. z vrcl. v $\tilde{M}$

1) $x_n \xrightarrow[n\to\infty]{d} \tilde{x}$ v $(\tilde{M}, d)$ zaplen $(M, d) \Rightarrow$

2) že $\tilde{M}$ zapl. v X

$X_1, X_2, \dots$ N EP z $E(X_1)=0$

$var(X_1) = \sigma^2 \rightsquigarrow E(X_1^2)=\sigma^2$

$var(X_1^2) = \tau^2$

a) kam konvergira

$$\sqrt{n}\left(\frac{\sum_{i=1}^{n} X_i^2}{n} - \sigma^2\right) =$$

$$\downarrow$$

$$\overset{S}{\parallel} \; E(X^2)$$

$$\frac{\sum X_i^2 - n\sigma^2}{\sqrt{n}}$$

$Y_1, Y_2, \dots$ NEP $\quad E(Y_1^2) < \infty$

Potem $S = \sum Y_i \quad \frac{S - E(Y_1)n}{\sqrt{var(Y_1)n}} \xrightarrow{d} N(0,1)$

$Y_i = X_i^2$

$var(X_i^2) = \tau_2$

$$\tau \frac{S - n\sigma^2}{\sqrt{\tau^2 n}} \longrightarrow \tau \cdot N(0,1) = N(0, \tau^2)$$

b) pokaži $\sqrt{n}\,\bar{X}^2 \xrightarrow{d} 0$

$$\bar{X} = \frac{X_1 + \dots X_n}{n} =$$

$\sqrt{n}\,\bar{X}\cdot\bar{X} \qquad \bar{X} \xrightarrow{d} 0 \leftarrow E(X_1)$

Po izreku o slabem zakonu velikih števil

Pot izreku slutskega

$\bar{X}\cdot(\sqrt{n}\bar{X})$

$$\sqrt{n}\,\bar{X} = \frac{\sigma X_1 + \dots + X_n - 0\cdot n}{\sqrt{n}\sigma} \xrightarrow{d} N(0,\sigma^2)(x)$$

$$\rightarrow \sqrt{n}\bar{X}^2 \xrightarrow{d} 0$$

c) $$M_n^2 = \frac{\sum (X_j - \bar{X})^2}{n}$$

kam konvergira $\sqrt{n}(M_n^2 - \sigma^2)$

$$\frac{\sum (X_j - \bar{X})^2 - n\sigma^2}{\sqrt{n}} =$$

$$= \frac{\sum X_j^2 - 2\sum X_j \bar{X} + \sum \bar{X} - n\sigma^2}{\sqrt{n}} \qquad \sum \bar{X} = n\bar{X}$$

$$\longrightarrow N(0,\tau^2)(x) + 0 + 0$$

Naj bodo $X_1, X_2 \ldots$ slučajne spremenljivke definirane na $(\Omega, \mathcal{F}, P)$. Pravimo da X_n konvergira k slučajni spr. X

a) v verjetnosti, če $\forall \varepsilon > 0$. $\lim_{n \to \infty} P(|X_n - X| > \varepsilon) = 0$

b) skoraj gotovo če $P(\lim X_n = X) = 1$

$$P(\{\omega \in \Omega \mid \lim X_n(\omega) = X(\omega)\}) = 1$$

c) $E(f(X_n)) \xrightarrow{d} E(f(X))$

$\mathbb{P} \to d$

$d \not\to \mathbb{P}$

$X(0,1)$ $\qquad \lim P(|X_n - X| > \varepsilon) =$

$X_n = -X$ $\qquad \lim (P(2|X| > \varepsilon) =$

$X_n \xrightarrow{d} X$ $\qquad \lim P(|X| > \frac{\varepsilon}{2}) = P(|X| > \frac{\varepsilon}{2})$

$\neq 0$ za $\varepsilon < 1$

$P \Rightarrow d$

$\forall \varepsilon > 0 \lim P(|X_n - X| > \varepsilon) = 0$

$E(\cancel{f(X_n)})$ f zwana, ograniczona pokrytka

$E(f(X_n))$ $\qquad I_n := \{ |X_n - X| > \varepsilon$

$E(f(X_n)) = E(f(X_n)(I_n + I_n^c)) =$

$= E(f(X_n) I_n) + E(f(X_n) I_n^c)$

$E(|f(X_n) - f(X)|) = E(|\ldots| \mathbb{1}_{I_n}) + E(|\ldots| \mathbb{1}_{I_n^c}) < c\varepsilon$

$\qquad\qquad\qquad\qquad \overset{c\varepsilon}{\vee}$

$\qquad\qquad\qquad\qquad \overset{\parallel}{0}$ $P(|X_n - X| > \varepsilon) = 0$

$X_n \xrightarrow{s.o} X$ če $P(\lim_{n\to\infty} X_n = X) = 1$

oziroma $P(\{\omega \in \Omega \mid \lim X_n(\omega) = X(\omega)\}) = 1$

$s.g \Rightarrow P$

$P \not\Rightarrow s.g$

$s.g \Rightarrow P$

zraven so lahko omeje

$$\{\lim_{n\to\infty} X_n = X\} = \{\forall \varepsilon > 0\ \exists N > 0,\ n \geq N \Rightarrow |X_n - X| < \varepsilon\}$$

$$\bigcap_{\varepsilon>0} \bigcup_{N\geq 0} \bigcap_{n\geq N} \{|X_n - X| < \varepsilon\} =$$

$$= \bigcap_{m\in\mathbb{N}} \underbrace{\bigcup_{N\geq 0} \bigcap_{n\geq N} \{|X_n - X| < \tfrac{1}{m}\}}_{A_m \text{ padajoče v } m}$$

$$= \bigcup_{N\geq 0} \underbrace{\bigcap_{n\geq N} \{|X_n - X| < \tfrac{1}{m}\}}_{B_N \text{ naraščajoče v } N}$$

Vemo:

$$P\left(\bigcap_{m\in\mathbb{N}} A_m\right) = 1$$

Vemo $\lim_{m\to\infty} P(A_m) = 1$

$$\lim_{m\to\infty} P\left(\bigcup_{N\geq 0} B_N\right) = 1 \qquad A_m \supseteq \bigcap A_m \qquad \Rightarrow P(\bigcap A_m) \leq P(A_m) \leq 1$$

$$\lim_{m\to\infty} \lim_{N\to\infty} P(B_N) = 1 \qquad B_N \subseteq \{X_N - X \mid < \tfrac{1}{m}\}$$

$$\lim_{N,m\to\infty} P(\{|X_N - X| < \tfrac{1}{m}\}) = 1$$

$P \neq S.g.$

$X_1(\omega) = \mathbb{1}_{[0,1]}(\omega)$

$X_2(\omega) = \mathbb{1}_{[0,\frac{1}{2}]}$ $\quad X_3 = \mathbb{1}_{[\frac{1}{2},1]}$

$X_4 = \mathbb{1}_{[0,\frac{1}{4}]}$ $\quad X_5 = \mathbb{1}_{[\frac{1}{4},\frac{1}{2}]}$, $\quad X_6 = \mathbb{1}_{[\frac{1}{2},\frac{3}{4})}$

$X = 0$

X_n :

Naj bo M in M' metrična prostora in $g: M \to M'$ preslikava s točkami (štetno mnogo) nezveznosti $D_g = \{x \in M \mid g \text{ ni zvezna v } x\}$

Naj bodo $x_1, x_2, \dots, x$ slučajni spremen. z vrednostmi v M in naj bo $P(X \in D_g) = 0$.

Če $X_n \xrightarrow[n\to\infty]{d,P,s,p} X \Rightarrow g(X_n) \xrightarrow[n\to\infty]{d,P,s,p} g(X)$